Does Electrocution Improve Roller-Crimper Cover Crop Termination?



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Introduction

- Organic systems cannot use herbicides, so cover crops must be killed mechanically **(Creamer & Dabney 2002)**
- The roller crimper (RC) reliably kills cereal rye only after anthesis; surviving tillers regrow **(Mirsky et al. 2009)**
- Electrical weed control (EWC) kills plants with high-voltage current, but one pass only top-kills mature rye **(Slaven et al. 2023; Sullivan 2025)**
- **Gap:** can EWC kill the tillers a crimper leaves alive, and does pass order matter?

Objectives

1. Test whether a same-day EWC pass, before or after roller crimping, improves cereal rye termination over either tool alone
2. Determine whether pass order (EWC→RC vs RC→EWC) changes termination and regrowth

Hypothesis: EWC + RC beats either tool alone, in either order.

Materials and Methods

5 TREATMENTS
Control · EWC · RC · RC→EWC · EWC→RC

COVER CROP
Cereal rye, Zadoks 85; ≈ 615 g m^{-2} biomass

EWC: ZASSO EH60
2.5 m, 3.2 km h^{-1}; 31–34 kW; 13.8–15.5 kJ m^{-2}

ROLLER CRIMPER
Commercial, 3 m wide; 1 pass, same day as EWC



Fig 1. EWC (Zasso EH60) and roller crimper in cereal rye.

Design: RCBD, 4 reps; College Station, TX, 2026

Measured: termination and survival (3–28 DAT), biomass, soil moisture

Analysis: ANOVA; Tukey's HSD (α = 0.05)

Results and Discussion



Fig 2. Plots 15 days after treatment (DAT): sequenced plots show the least regrowth.



Fig 4. Proposed mechanism (conceptual): EWC top-kills, the crimper misses tillers; together, few survive.

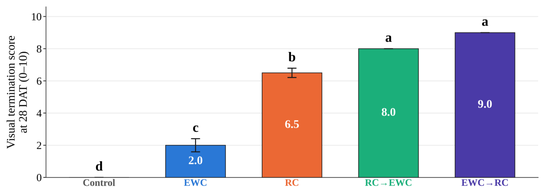


Fig 3. Both sequences terminated rye best (8–9 of 10) at 28 DAT; order did not differ.

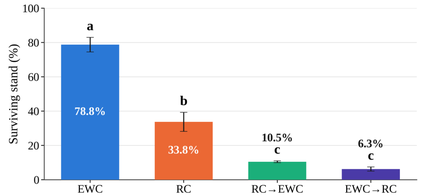


Fig 5. Only 6–11% of rye survived the sequences, vs 34–79% with one tool.

Conclusions

- **EWC + RC beat either tool alone:** termination rose from 65% to 80–90%; surviving rye fell from 34% to 6–11%
- **Pass order did not matter** (P = 0.08): EWC can run before or after crimping
- **Sequences cut biomass 57–62%;** crimped plots held 27–28% soil moisture vs 17% in the control **(Kornecki et al. 2012)**
- **EWC alone failed** even at ≈ 10× the dose that killed an oat–clover mix **(WSU 2025)**

TAKEAWAY
Add a same-day EWC pass to roller crimping, in either order.

Future Research

- Terminate earlier (anthesis) and test multiple EWC passes
- Test other cover crops, weed suppression and cash-crop yield

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